The Leader in IoT Wireless Connectivity

100%
IoT Focused

amazon sidewalk · Bluetooth · matter · Multiprotocol · Proprietary · THREAD · Wi Fi · Wi SUN · zigbee · Z WAVE

Breadth and Depth of Wireless IoT Protocols

#1
Share in Mesh

1st
To Market with Multiprotocol, BLE Mesh, BLE 5.1

Innovation
Performance, Power, CoEx, Modules, SecureVault™

ember	ENERGY micro	bluegiga	telegesis	Micriμm	ZENTRI	Z WAVE	REDPINE SIGNALS
2012	2013	2015	2015	2016	2017	2018	2020
Software ZigBee SoC	Low-power 32-bit MCUs	BT Smart Modules	ZigBee/Thread Modules	Software RTOS	Cloud Connected Wi-Fi	Smart Home Protocol	Ultra Low Power Wi-Fi

A Complete and Robust Bluetooth® Offering

DESIGN | DEVELOPMENT | DEPLOYMENT

Developer Kits

- Exploratory kits with standard connectors
- Full featured kits with LCD, Power Measurement, etc.
- Demo kits w/ built in sensors

200+ OOB demos and sample applications

Cutting-edge BT Stack

- Bluetooth LE and Mesh
- Supporting most BLE features
- Certified against latest spec

~1B BLE units deployed

Mobile Apps

- BLE scanner/advertiser
- Demos
- IOP testing with smartphones
- Bluetooth Mesh ADK and App

100+ Mobile phones with next-gen IOP testing

ICs and Modules

- QFN, CSP ICs
- PCB and SiP modules
- Antenna Design (int/ext)
- Optimized size
- Pre-certified modules

1st in the market with PSA L3 certification

Superior Developer Tools

- GATT Configurator
- Bluetooth NCP Commander
- Network Analyzer
- AoA Analyzer
- Energy Profiler

~1M Packets transmitted daily for Mesh endurance testing

Reliability Testing, IOP

- Large Scale Test Network
- IOP testing
- Regression testing & CI
- Throughput and range test

7,000+ functional tests for Bluetooth and mesh

2025

- SiBG301, first Series 3 BLE product, announced
- BG29 announced
- BG22L and 24L launched to boost BLE portfolio
- Channel Sounding Dev Kit Launched

2024

- BT 6.0 spec adoption
- ESL Large-Scale Test Network (~1000+ nodes)
- BG26 announced
- Channel Sounding support available

2023

- BT 5.4 spec announced
- SiLabs adoption of 5.4
- BG27 released with 5.4
- Mesh Large Scale Network

2022

- BG24 released w/ 5.3
- Mesh 1.1 announced
- Series 2 SoCs support mesh 1.1

2021

- First PSA L3 SoC
- BT 5.3 spec adoption
- Enabled CPMS integration
- Series 2 SoCs support 5.3

2015

Shipped 1st BLE SoC & module

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Digital Twin of Solar Assets

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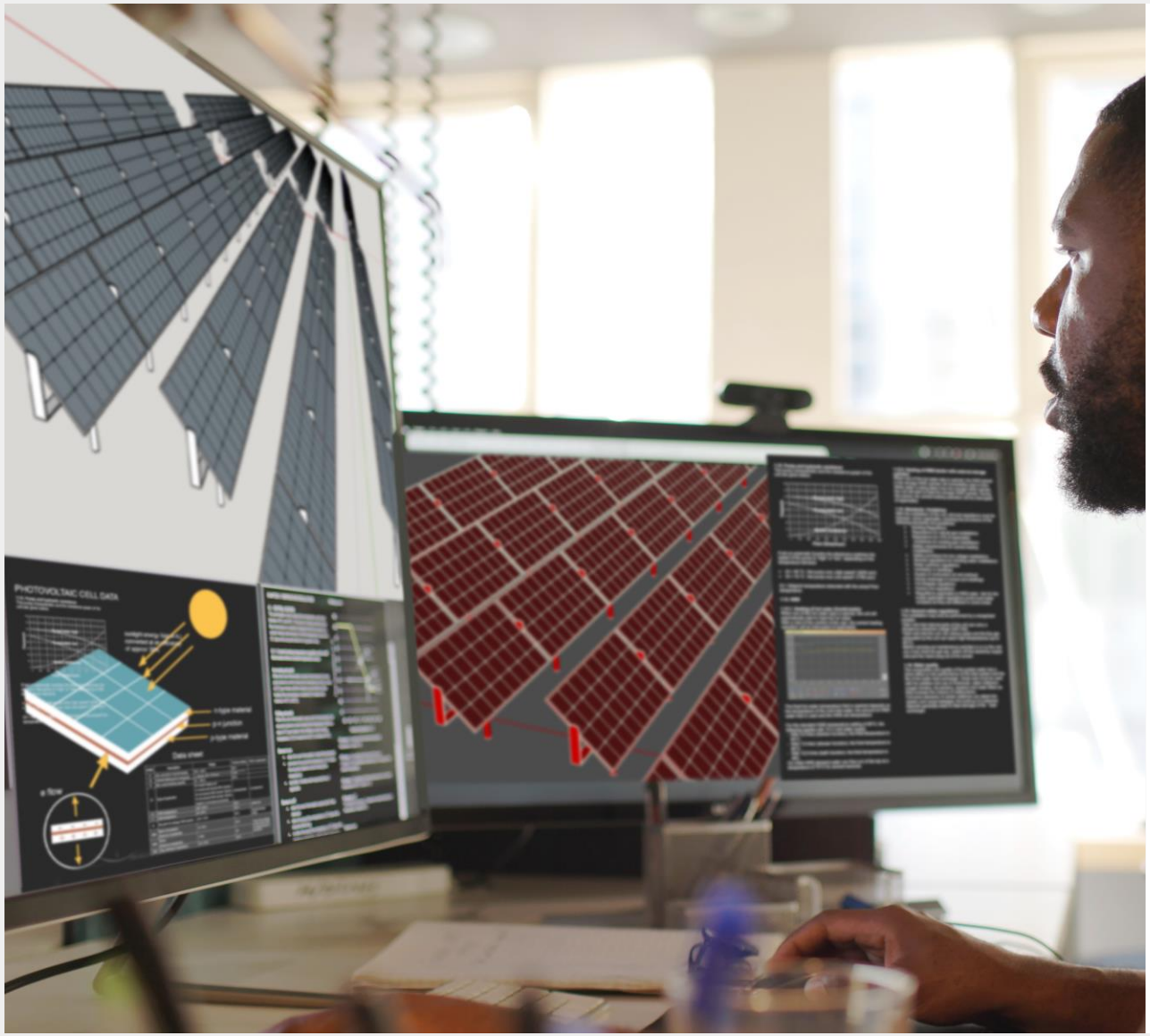
Digital Twin of Solar PV Deployments



Model: A virtual representation of the solar farm.

Live Data: Real-time information on the solar panels and equipment.

Predictions: Tools to forecast issues and maintenance needs.

History: Past performance and maintenance records.

Simulations: Ability to test different scenarios and their impacts.

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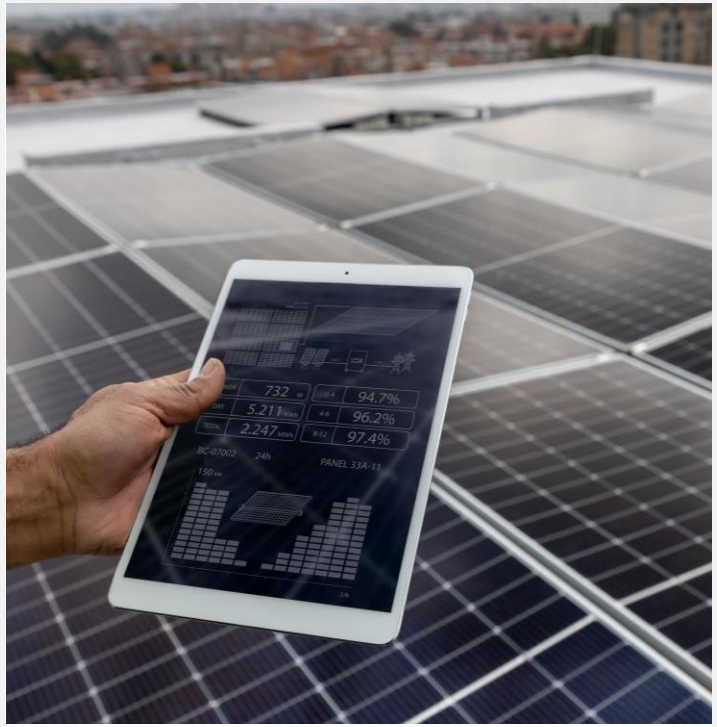
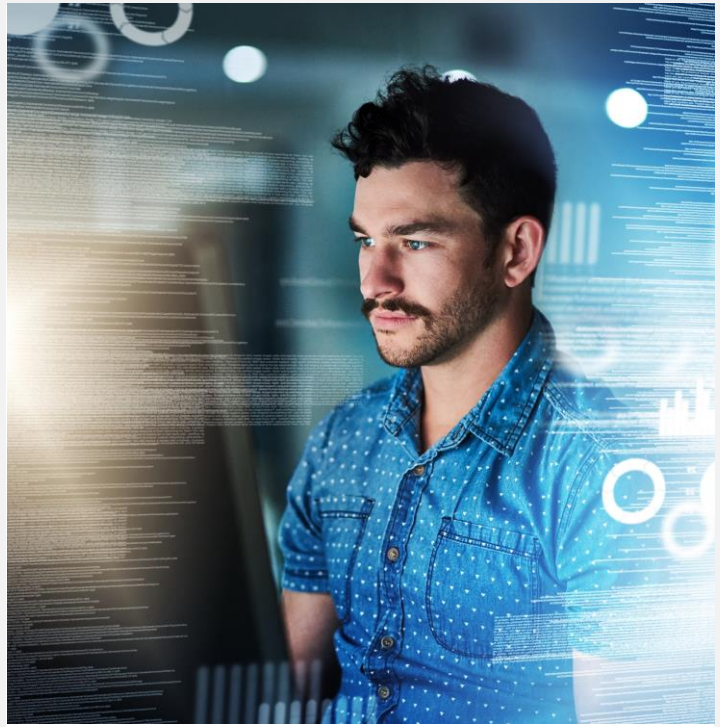
Digital Twin of Solar PV Deployments





COMMISSION

Device onboarding, ranging-based mapping to automatically create a model of the installation

Bluetooth Channel-Sounding

MONITOR

Live view of energy production and performance. Edge intelligence

Wi-SUN, IEEE 15.4, Wi-Fi, Wired

ANALYZE

Simulations, predictions, actions and historical performance at the backend

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Use Cases for Digital Twin

Bluetooth

Accurate Drone inspections

Worker guidance

Mapping

Wi-SUN / 15.4 / Wired

Sensing

Power / Energy

Control

Analytics, Supervisory Control, Historical Data, Grid Integration

Efficiency, Power Output, Temperature Coefficient, Degradation Rate, Tilt / Orientation, Weather, Forecasts

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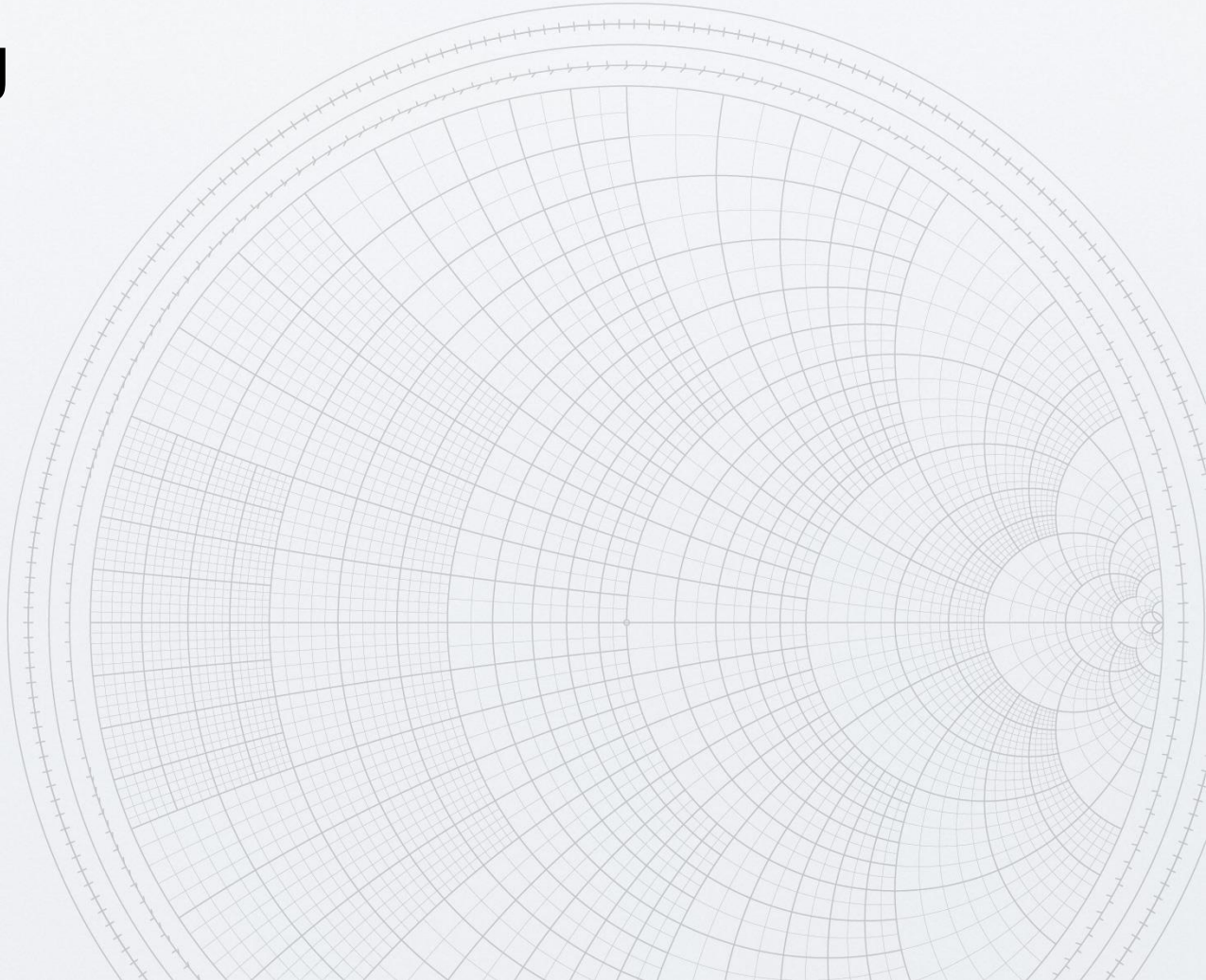
Bluetooth® Channel Sounding for Solar Commissioning and Automapping



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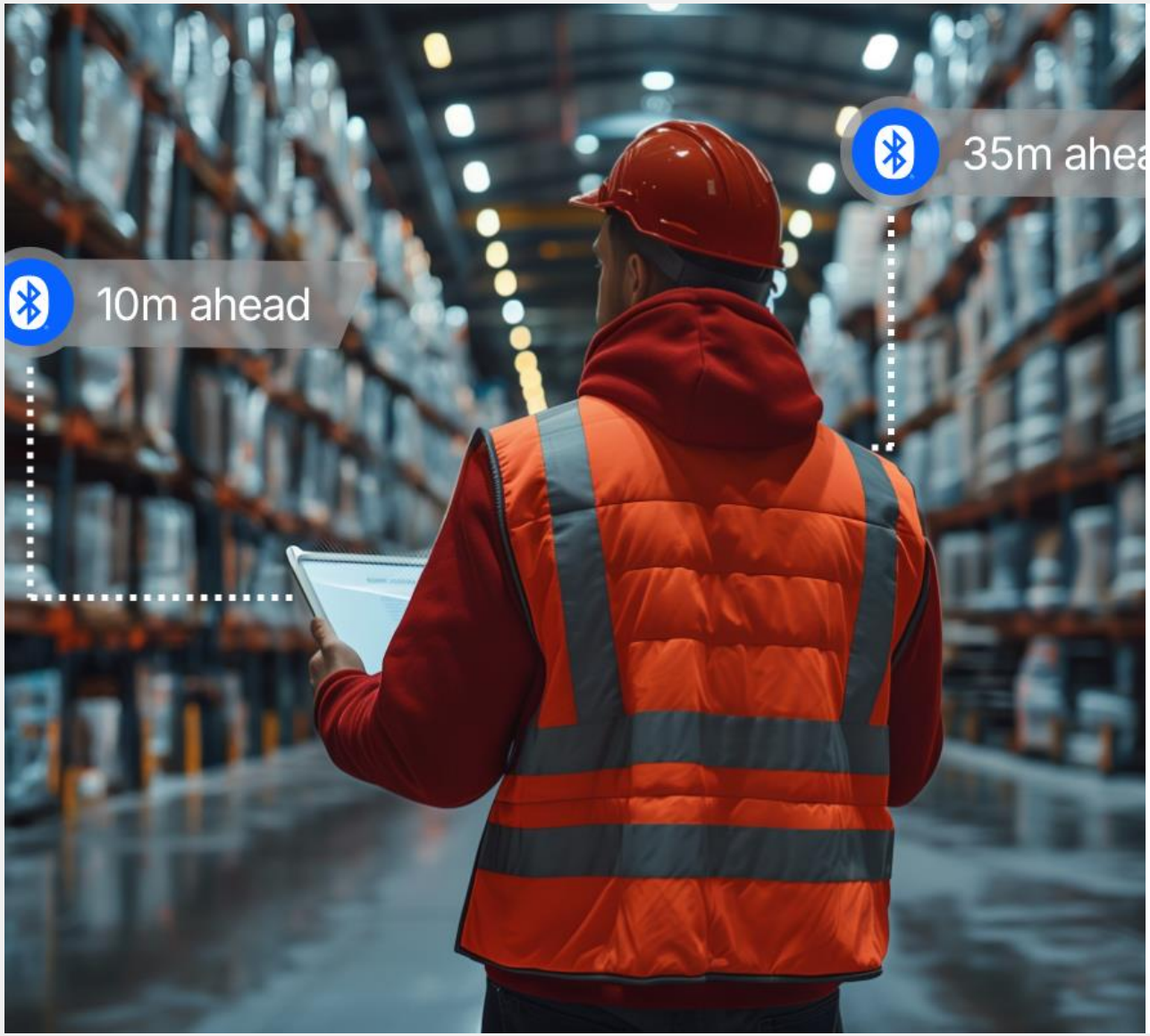
Driving Innovation with Bluetooth® Channel Sounding



Enables high-accuracy distance measurement

- Resolve distance between two connected Bluetooth LE devices
- Support for phased based ranging technique and Time of Flight
- Multi-antenna supported for optimized Non-Line-Of-Sight ranging

Standardized for interoperability and reliability

- Ensures compatibility with a wide range of devices
- Device performance ensured through Bluetooth qualification

Cost effective and efficient

- Unlike UWB based solutions, potential for single-chip ranging
- Simplifies layout design, saving board space and cost

Enhanced Security

- Reduces likelihood of spoofing and man in the middle attacks
- Deterministic random bit generator randomizes the bit stream, payloads in tone extensions, antenna path selection and more
- NADM, or normalized attack detector metric, communicates likelihood of bit manipulation in RTT
- Mode3 (RTT+PBR) allows cross-check of data from PBR with RTT to prevent man-in-the-middle (MITM) attack

Flexible architecture

- Software stack supports device operating in NCP or SoC mode

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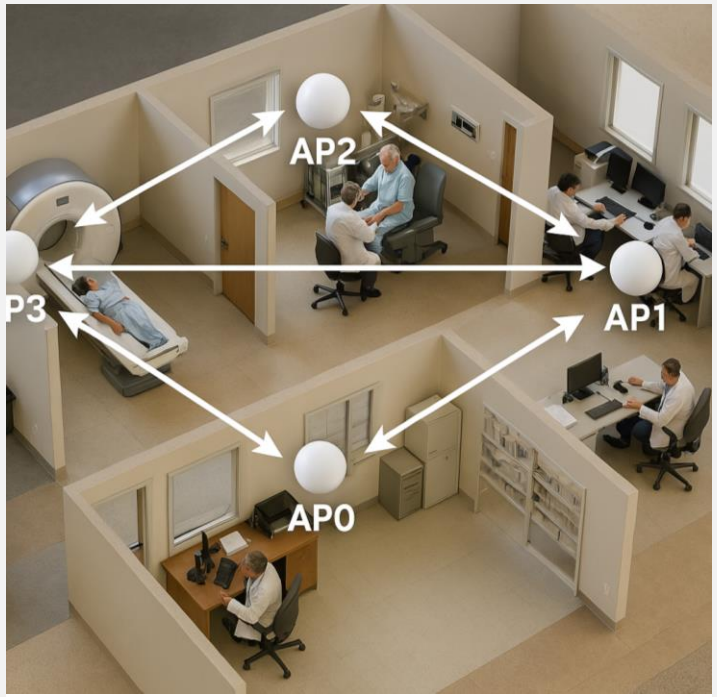
Bluetooth Channel Sounding - Target Markets & Use Cases

PROXIMITY AWARENESS

Door locks
Keyless entry
Building access systems
Geofencing - security alerts

LOCALIZATION

Indoor asset management - hospitals, warehouses
Pet tracking inside home
Item finding - wallet, keys



AUTOMAPPING

Luminaires, Access Points
Solar Trackers
Solar Optimizers and Microinverters
Accurate Mapping for Battery Storage Units

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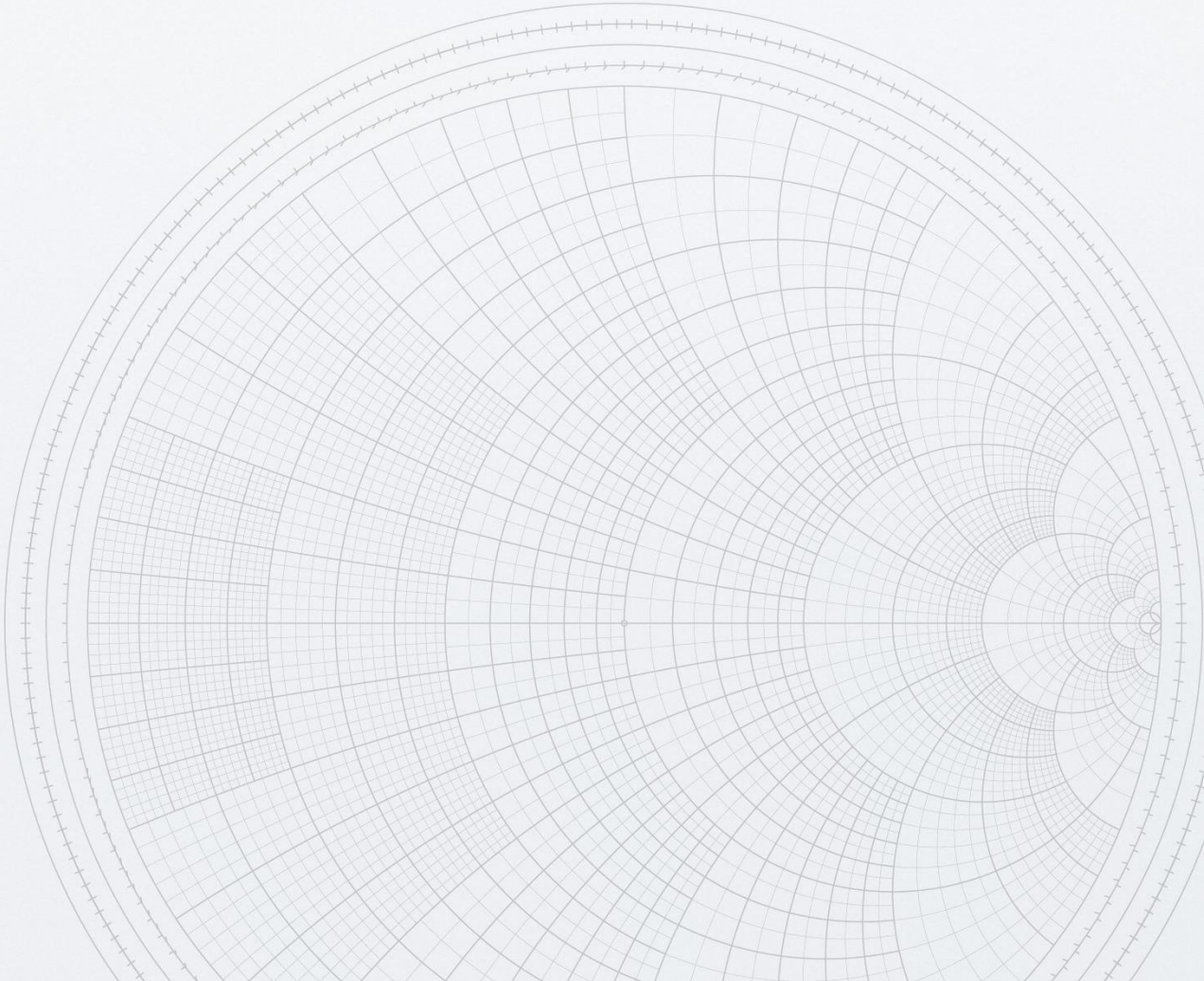
Channel Sounding for Automapping



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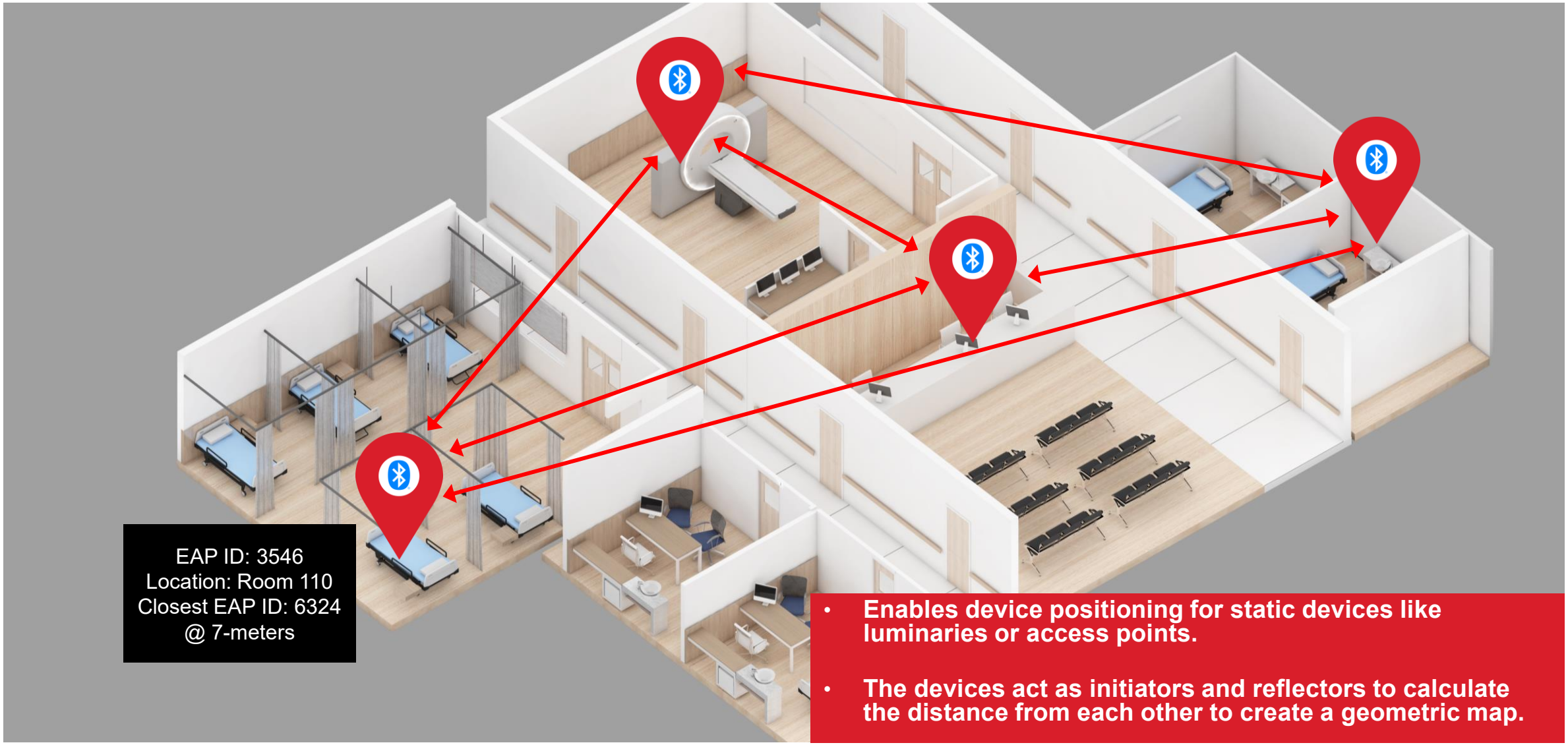
Channel Sounding for Static Device Positioning



- **Enables device positioning for static devices like luminaries or access points.**
- **The devices act as initiators and reflectors to calculate the distance from each other to create a geometric map.**

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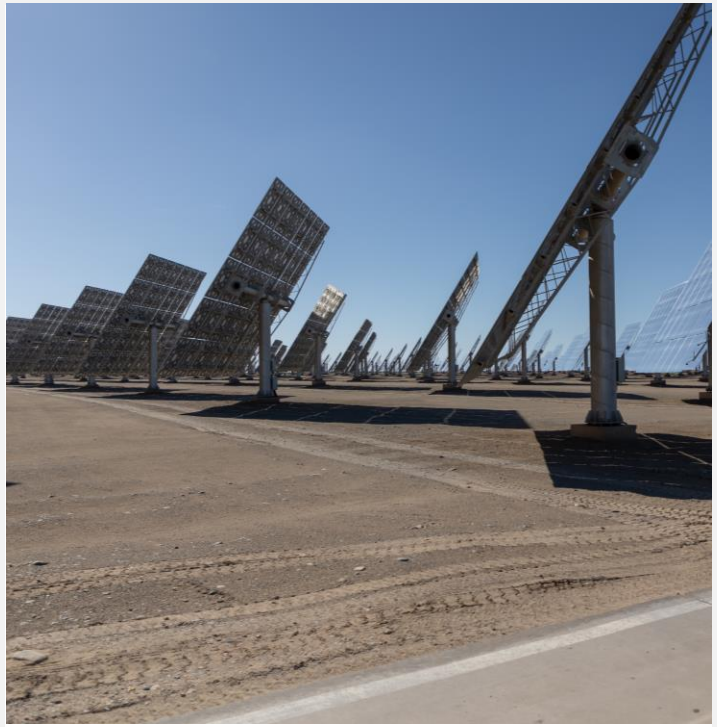
Device Types in Renewable Energy to Commission and AutoMap

MODULE LEVEL POWER ELECTRONICS

Few to several thousand devices

Shortest distance 1m



SOLAR TRACKERS

Several Hundred devices

Shortest distance several meters

ENERGY STORAGE

Few to several dozen devices

Shortest distance 25cm (modules) to several meters (containers)

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Channel Sounding Automapping in Solar and Storage

Key value propositions

- Significantly reduces labor required for commissioning and device mapping
- Supports workflows where installation & commissioning are handled by different personnel
- Ideal for challenging environments, including hard-to-access rooftop installations

Use Case details

- Powered

  Devices have a stable power source
- Not timing critical

  Ranging can be performed without strict timing requirements
- Stationary assets

  Devices remain fixed during ranging/measurement phase
- Known orientation

  Physical orientation is predetermined and consistent
- External antennas acceptable

  No major constraints on antenna form factor
- Backend connectivity available

  Wired or wireless options exist
- No continuous ranging

  Used during commissioning or on-demand only

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Module Level Power Electronics (MLPE)

Use case: Automap installed MLPEs

- Reduces the needed human intervention
- Avoid crosstalk with neighboring systems

Installations

- Residential (1 to 10s of nodes)
- Commercial (10 to 1000+ nodes)
- Utility (1000 to 10000+ nodes)

Ranging requirement

- 1 MLPE per panel
- Panel sizes: 1 – 1.5m x 1.8 – 2.5 m

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Solar Trackers

Use Case: Automap installed trackers when commissioning the solar farm

Ranging requirements

- Tracker every few panels to full row
- Range between trackers 3m – tens of meters

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Battery Storage

- **Use cases:**
  - **Map and locate battery modules in racks**
  - **Mapping for Utility Energy Storage in fields**

- **Ranging requirements**
  - Battery modules in Stacked style battery (rack)
    - Dimension example
      10” high (25cm)
      20” long (50cm)
      12” deep (30cm)
  - Utility ESS can have several packs stacked and multiple columns

  - Utility ESS
    - Container size e.g. 6.0m x 2m up to 12m x 2m
    - 1 GWh system use 500–1,000 containers depending on container size/capacity

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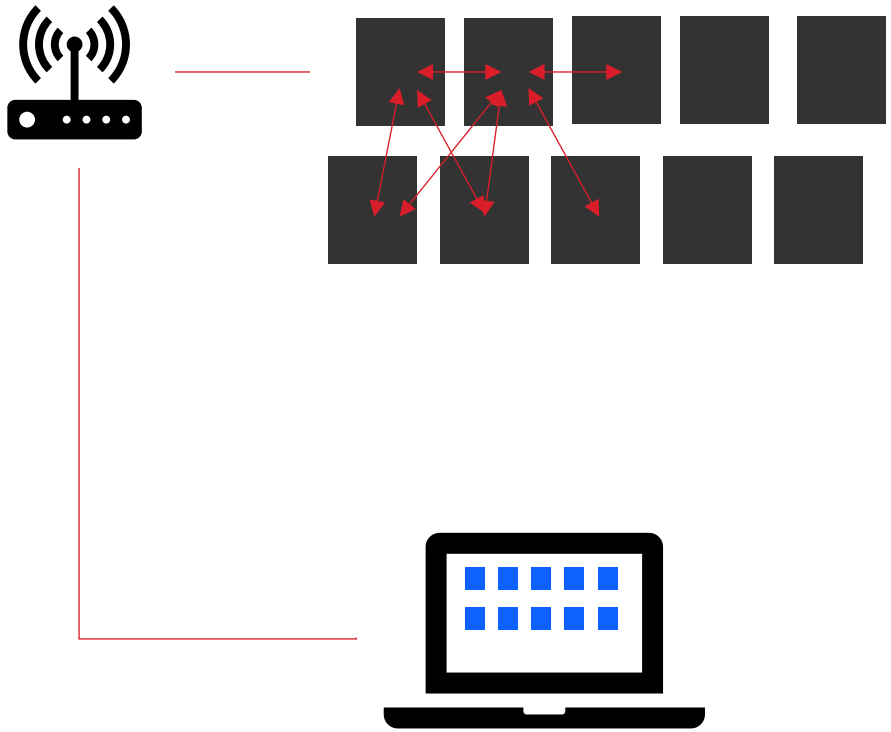
Concept: Channel Sounding for Solar Commissioning and Automapping



- **Using Channel Sounding to create a relative map of PV assets**
  - First MLPE to start ranging to its closest neighbors
    - Establishes connections, performs ranging, disconnects and repeat
  - Followed by other panels in sequence
  - IDs of assets & relative distance sent to backend for mapping
- **CS PBR offers robust ranging between two assets using Bluetooth connections**
- **+/- 0.5m accuracy sufficient to build a map**
- **Not timing critical use case – round-robin until all relative measurements taken**
  - <50ms per over the air measurements + connection overhead and calculation

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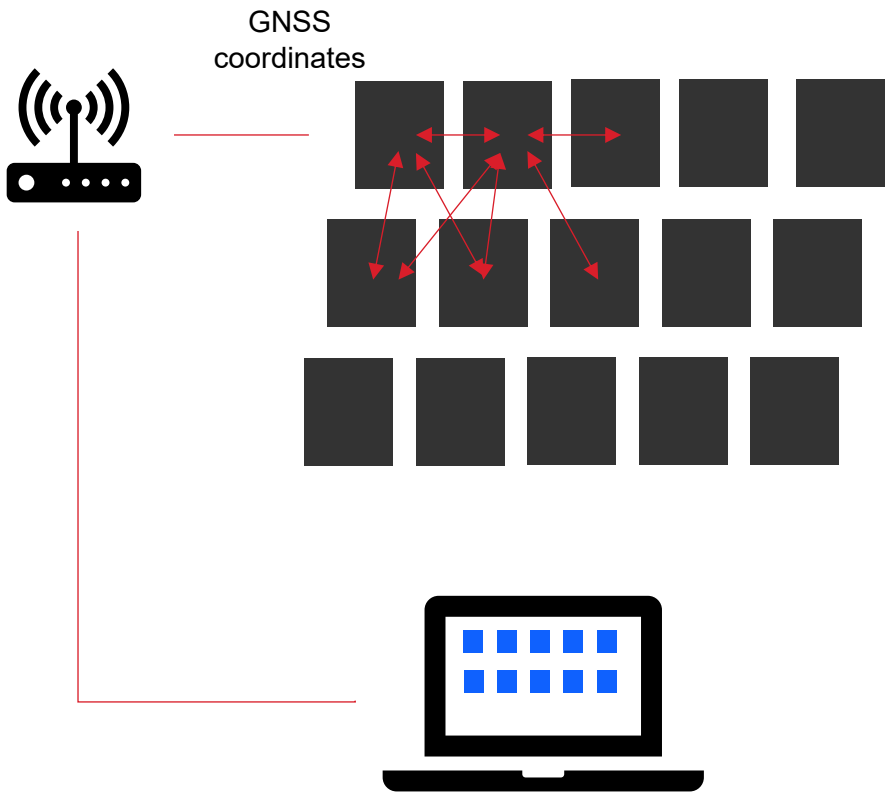
Concept: Channel Sounding for Solar Commissioning and Automapping



- **Anchor point(s) with GNSS coordinates act as starting point and reference**
- **Relative distance between neighboring nodes is measured**
  - Measurements trickle down as a wave
- **Data sent to backend for mapping**
  - Connectivity not limited to Bluetooth and OoB options can be used. (e.g. Mesh Technoglogies Wi-SUN, IEEE15.4, wired)

Hypothetical calculations

- **Ranging Measurements (with immediate neighbor):**
  - 100 nodes (10x10 grid): 342
  - 1000 nodes (100x10 grid): 3672
  - 10000 nodes (100x100 grid) 39402

- **Ranging measurement**
  - 0,3s (Con inv 50ms, 72ch, 4 antenna paths, CS inv 300ms)
  - Ranging calculation 180ms at initiator
  - 5 ranging samples
  - 2 sec per range estimation between two panels

- **100 nodes: 11min**
- **1000 nodes: 2h**
- **10000 nodes: 22h**

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Silicon Labs Automapping Demo

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Automapping Demo

- **Requirements**
  - Hardware requirements
    - 4-5 pcs BRD2606A boards, battery powered
    - PC with Silabs Bluetooth dongle/devkit
  - SW requirements
    - PC with Python (pyBGAPI)
    - BRD2606 with SiSDK2024.12.1 or later

- **Operation**
  - Multiple battery powered BRD2606A devices placed at different locations in a room.
  - Python application running PC with Bluetooth dongle
  - Wirelessly orchestrates via Bluetooth interface ranging between each device pairs
  - Retrieves ranging results wirelessly
  - Dynamically builds and plots geometric map based on the ranging results on the PC SW

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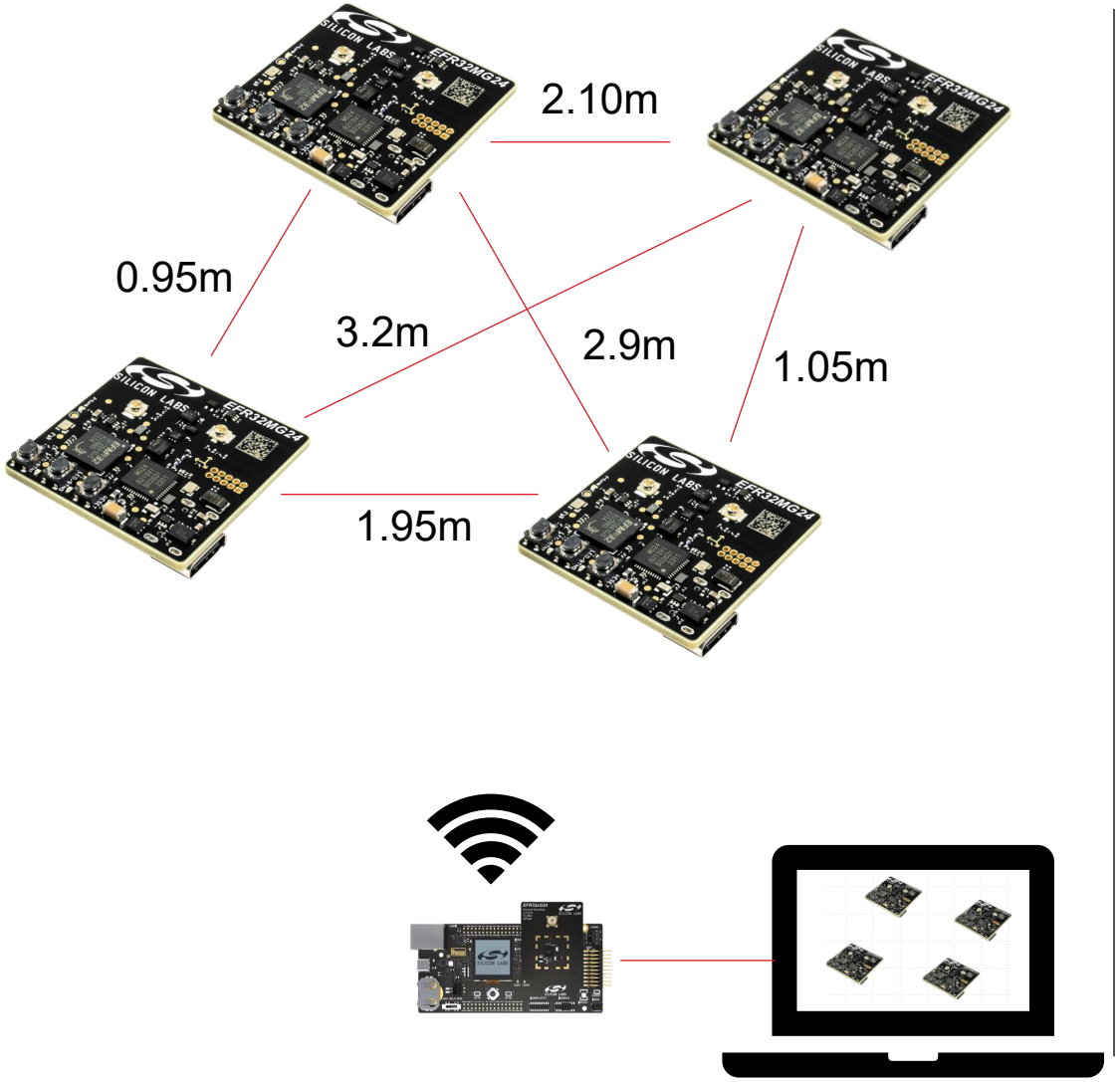
Automapping Demo Details



- **Optimized for xG24-DK2606A**
- **Continuous mapping or one-shot**
- **Each node act as Initiator and Reflector**
- **PC connected node orchestrates the roles and collects the ranging results**
- **PC Application used to build the map**
- **Performance**
  - Time to AutoMap – currently 2 min
    - Function of accuracy and responsiveness
    - A lot of variables – connection interval, No. of CS channels used, CS algorithm used, etc.
  - Accuracy
    - NLOS <1.5m
    - LOS <0.5m

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Static Device Positioning Demo

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Maximize Returns— Automate Commissioning of Solar Assets

EVALUATE

- **Order xG24-DK2026A**
  - Minimum 4 kits for meaningful mapping
  - Can be extended as many as needed
- **Ask Silicon Labs representative for Automapping demo SW**
  - Includes prebuilt binaries
  - Host code for Python
  - Source codes for future development

DESIGN HW AND EMBEDDED SW

- **Design HW using EFR32MG24 or EFR32MG26**
  - MLPE, Trackers, other
- **Consider EFR32 for other functionality**
  - Rapid-shutdown
  - Monitor and control over wireless
  - MCU (MPPT, Motor control)
- **Distance algorithm from Silicon Labs**

DESIGN BACKEND SERVICES

- **Build automapping functionality to commissioning tool**
  - Distance measurements from asset to build relative map between nodes
  - Attach GNSS location or roof plan information relative map
- **Add digital twin use cases with distance information**

Ask Silicon Labs representative for design consultants and partners accelerated Time-to-Market!

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BG24: Optimized for Battery Powered, Channel Sounding-enabled IoT Devices

Bluetooth® Proprietary

- **5x5 QFN40 (26 GPIO), AEC-Q100**
- **6x6 QFN48 (32 GPIO), AEC-Q100**
- **3.1x3.0 WLCSP42**

DIFFERENTIATED FEATURES

- **Ultra small form-factor**
  - 3.1 x 3.0 WLCSP package
- **+20 dBm output power**
  - Eliminates need for external power amplify
- **AI/ML accelerator**
  - Accelerates inferencing while reducing power consumption
- **Secure Vault High**
  - Protects data and device from local and remote attacks
- **20-bit ADC**
  - 16-bit ENOB for advance sensing
- **Improved Coexistence**
  - Ideal for gateways and hubs
- **PLFRCO**
  - Eliminates need for 32 KHz xtal

DEVICE SPECIFICATIONS

- **High Performance Radio**
  - Up to +19.5 dBm TX
  - -97.6 dBm RX @ BLE 1 Mbps
- **Efficient ARM® Cortex®-M33**
  - Up to 78 MHz
  - 1536kB Flash, 256kB RAM
- **Low Power**
  - 49.1 µA/MHz (CoreMark)
  - 5.0 mA TX @ 0 dBm
  - 5.1 mA RX (802.15.4)
  - 4.4 mA RX (BLE 1 Mbps)
  - 1.3 µA EM2 sleep
- **Multiple protocol support**
  - Bluetooth 5.4 (1M/2M/LR), Bluetooth mesh, Proprietary 2.4 GHz

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BG24L: Channel Sounding Optimized, High-Performance & Low-Cost AI/ML Wireless SoC

Bluetooth®

- **5x5 QFN40 (26 GPIO)**

DIFFERENTIATED FEATURES

- **Supports Bluetooth 6.0**
  - Channel Sounding optimized BLE SoC
  - Single-connection two-way ranging
  - Ideal Solution for Channel Sounding tags
- **Lowest Power RF**
  - Increases battery life
- **PLFRCO**
  - Eliminates need for 32 KHz XTAL and lowers overall system cost
- **16-bit ADC**
  - Up to 14-bit ENOB for better analog sensing
- **AI/ML accelerator**
  - Accelerates inferencing while reducing power consumption
- **Secure Vault Mid**
  - Protects data and device from local and remote attacks
- **Improved Coexistence**
  - Ideal for gateways and hubs

DEVICE SPECIFICATIONS

- **High Performance Radio**
  - Up to +10 dBm TX
  - -97.6 dBm RX @ BLE 1 Mbps
- **Efficient ARM® Cortex®-M33**
  - Up to 78 MHz
  - 768kB Flash, 96kB RAM
- **Low Power**
  - 49.1 µA/MHz (CoreMark)
  - 5.0 mA TX @ 0 dBm
  - 5.1 mA RX (802.15.4)
  - 4.4 mA RX (BLE 1 Mbps)
  - 1.3 µA EM2 (16kB RAM retention)
- **Wide Operating Range**
  - 1.71 to 3.8 volts
  - +125ºC operating temperature
- **Multiple protocol support**
  - Bluetooth 6.0 (1M/2M/LR), Bluetooth mesh, Proprietary 2.4 GHz

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Dual-Antenna Channel Sounding board

- **xG24 Channel Sounding Development Kit**
  - Available in Feb 2025
  - Development Kit with two PCB antennas
    - Antenna diversity offers increased robustness and accuracy
  - Intra-event antenna switching for optimal non-line of sight performance
  - Includes IMU sensor to detect movement & wake-up the tag
  - Small form factor
    - Ideal for size-constrained applications like key fobs
  - AEC-Q100 Compliant
  - SoC/NCP Sample Apps
    - Initiator and Reflector examples supported
  - Ranging Library
    - Process IQ samples, post-filtering, and compute distance using configurable algorithm

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End-to-End Channel Sounding Solution: Integrated, In-House, Royalty-Free

- **Channel Sounding Algorithm Features**
  - Supports Multiple Channel Sounding Ranging Modes
    - PBR, RTT, PBR with RTT as submode
  - Antenna Switching
    - Built-in support for antenna diversity. Supports 1, 2 and 4 antenna paths
  - Supported Algorithm modes
    - Static mode – High accuracy
    - Real Time Basic – Offers high accuracy with increased computation and measurement latency; supports tracking speeds up to 1 m/s
    - Real Time Fast – Offers moderate accuracy with lower computation and measurement latency, supports tracking up to 2.1 m/s
  - Configurable Channel Selection (72, 37, or 20 Channels)
    - Selectable based on accuracy needs and power constraints
  - Integrated Real-time visualization tool with Simplicity Studio
    - Object tracking – Displays accurate range to the target device
    - Relative Radial Velocity metric – Calculates how quickly devices move toward or away from each other

- **Key Benefits Of Our In-House Solution**
  - Licensing-Free Deployment
    - Eliminates third-party royalties, simplifying BOM cost structure and legal risk
  - Optimized HW-SW Co-Design
    - Tight coupling between silicon & firmware ensures seamless performance & efficiency
  - Single-Vendor Lifecycle Support
    - Unified hardware & software ownership streamlines debugging, validation, and updates

Feature	Silicon Labs	Closest Competitor
Algorithm IP	✅ In-house developed	❌ Licensed 3rd-Party
Software Stack	✅ Integrated FW + SW	⚠️ Layered Integration with External IP Blocks
IP Licensing Model	✅ Fully Owned Stack, Royalty-Free	❌ Perpetual or Usage-Based Licensing Fees
Vendor Support	✅ Unified Silicon + Stack Support	⚠️ Fragmented Support Across Vendors

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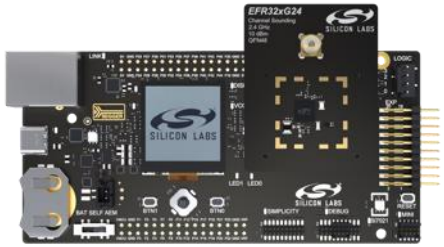
Silicon Labs Channel Sounding – Complete Offering



DEVELOPMENT KITS

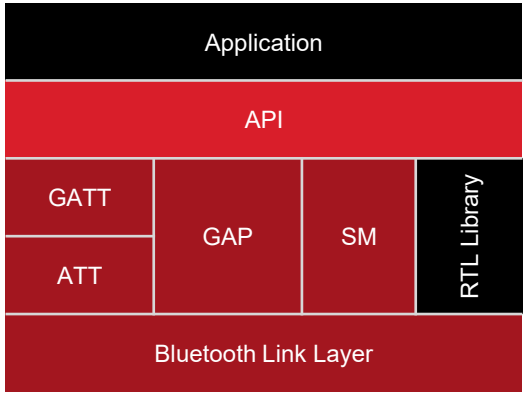
BRD4198A with Single Antenna

Wireless Pro Kit

EFR32MG24 + 10dBm OPN

BRD2606A with Dual Antennas

Bluetooth®



STACK SOFTWARE

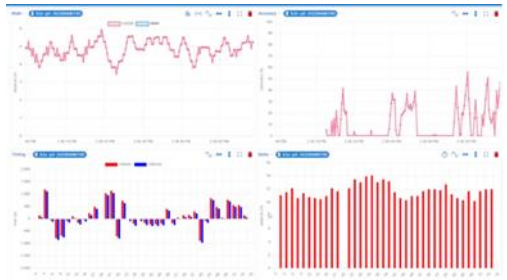
In-house developed stack

Supports Bluetooth 6.0 features + Channel Sounding

New and improved Ranging features

Si

SIMPLICITY STUDIO



DEVELOPMENT TOOLS

Real-time visualization tool
PBR, RTT modes
CS Sample projects
CS Analyzer + Energy Profiler + Network Analyzer
App note + Salesforce Support

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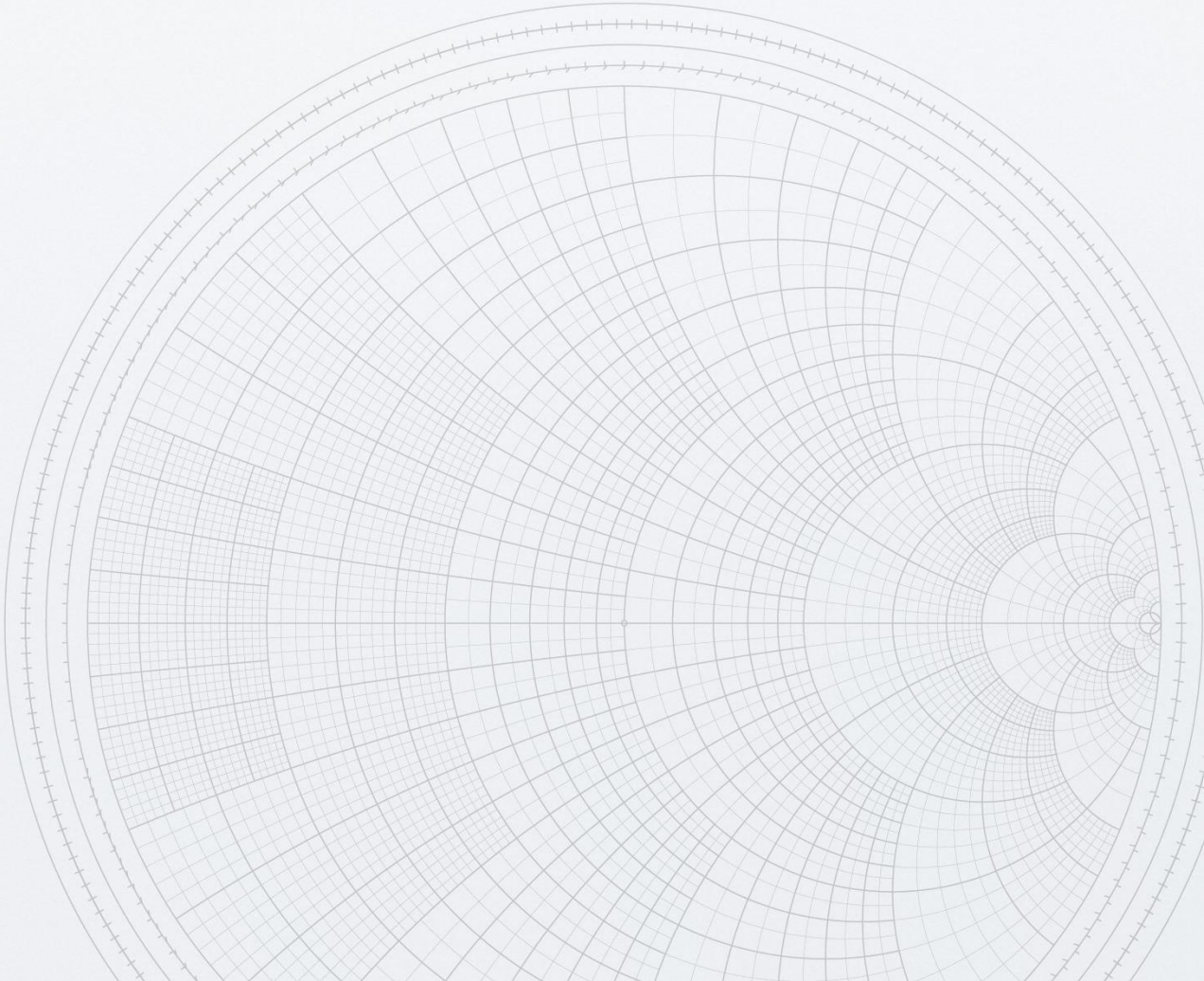
Thank You



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